

# The Ensemble theory And applications

in quantum statistical mechanics

## Introduction

In conventional Statistical Mechanics probabilities, expectation values, variances and higher moments of observables are computed via ensemble averages. Depending on the situation under consideration one must employ the microcanonical, canonical or the appropriate grand canonical ensemble. The validity of this approach is beyond all doubt and the results obtained using it have been confirmed by innumerable experiments. On the other hand, the role of probability in Physics, the problem of ergodicity and especially the microscopic justification of the Second Law of Thermodynamic are very subtle issues and many fundamental questions concerning them are still open despite many decades of research[1]. The starting point of our discussion will be to show how the applicability of ensemble averages can be justified using Quantum Mechanics and measure concentration techniques without any extra assumptions.

## Ensembles Definition

As a system is defined by the collection of a large number of particles, so the “ensembles” can be defined as a collection of a number macroscopically identical but essentially independent systems. Here the term *macroscopically identical* means, as, each of the systems constituting an ensemble satisfies the same macroscopic conditions, like *Volume, Energy, Pressure, Temperature and Total number of particles etc.* Here again, the term *essentially independent* means the system (in the ensemble) being mutually non-interacting to others, i.e., the systems differ in microscopic conditions like *parity, symmetry, quantum states etc*[2].

There are three types of ensembles:

1. Micro-canonical Ensemble
2. Canonical Ensemble
3. Grand Canonical Ensemble

## Micro-canonical Ensemble

It is the collection of a large number of essentially independent systems having the same energy  $E$ , volume  $V$  and total number of particles  $N$ .

The systems of a micro-canonical ensemble are separated by rigid impermeable and insulated walls, such that the values of  $E$ ,  $V$  &  $N$  are not affected by the mutual pressure of other systems.

This ensemble is as shown in the figure below[3].

Here all the borders are impermeable and insulated

|                                                                         |                                                                         |                                                                         |                                                                         |                                                                         |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|
| System 1<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 2<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 3<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 4<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 5<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$  |
| System 6<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 7<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 8<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 9<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 10<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ |
| System 11<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 12<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 13<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 14<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 15<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ |
| System 16<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 17<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 18<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 19<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 20<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ |
| System 21<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 22<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 23<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 24<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 25<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ |
| System 26<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 27<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 28<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 29<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 30<br>Energy $E$<br>Volume $V$<br><br>Number of<br>Particles $N$ |

All the walls here are  
rigid, impermeable  
and insulated.

## Canonical Ensemble

It's the collection of a large number of essentially independent systems having the same temperature  $T$ , volume  $V$  and the number of particles  $N$ .

The equality of temperature of all the systems can be achieved by bringing all the systems in thermal contact. Hence, in this ensemble the systems are separated by rigid impermeable but conducting walls, the outer walls of the ensemble are perfectly insulated and impermeable though[3].

This ensemble is as shown in the

|                                                                        |                                                                        |                                                                        |                                                                        |                                                                        |
|------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|
| System 1<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 2<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 3<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 4<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 5<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$  |
| System 6<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 7<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 8<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 9<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$  | System 10<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ |
| System 11<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 12<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 13<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 14<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 15<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ |
| System 16<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 17<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 18<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 19<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 20<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ |
| System 21<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 22<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 23<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 24<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 25<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ |
| System 26<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 27<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 28<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 29<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ | System 30<br>Temp. $T$<br>Volume $V$<br><br>Number of<br>Particles $N$ |

Outer walls here are  
rigid, impermeable  
and insulated.

Inner walls are  
rigid, impermeable  
but conducting.

## Grand Canonical Ensemble

It is the collection of a large number of essentially independent systems having the same temperature  $T$ , volume  $V$  & chemical potential  $\mu$ .

The systems of a grand canonical ensemble are separated by rigid permeable and conducting walls[3]. This ensemble is as shown in the figure

|                                                                      |                                                                      |                                                                      |                                                                      |                                                                      |
|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|
| System 1<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$  | System 2<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$  | System 3<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$  | System 4<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$  | System 5<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$  |
| System 6<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$  | System 7<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$  | System 8<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$  | System 9<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$  | System 10<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ |
| System 11<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 12<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 13<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 14<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 15<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ |
| System 16<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 17<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 18<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 19<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 20<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ |
| System 21<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 22<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 23<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 24<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 25<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ |
| System 26<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 27<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 28<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 29<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ | System 30<br>Temp. $T$<br>Volume $V$<br><br>Chemical Potential $\mu$ |

Outer walls here are rigid, impermeable and insulated.

Inner walls are rigid, permeable and conducting.

## How the ensembles are related

the names “microcanonical”, “canonical” and “grand canonical” suggest a hierarchy of ensembles. This is intentional. Suppose that we have a very large grand ensemble consisting of a number of identical volumes which communicate with each other. Imagine that we simultaneously shut all the doors connecting these volumes. The number of molecules in each volume is now fixed. We can extract from the original grand ensemble a set of canonical ensembles by collecting together all the volumes which have the same value of  $N$ . In a sense then, a grand ensemble is an ensemble of canonical ensembles. We can see this relationship in the form of the grand partition function (equation 6): It is an average of canonical partition functions weighted by a function of  $N$ . Similarly, suppose that we have a very large canonical ensemble consisting of a number of identical replicas in a thermostat device at temperature  $T$ . If we simultaneously isolate all the replicas, we now have a number of systems each of which has a fixed value of  $U$ . We can make microcanonical ensembles by collecting together all the members of the original canonical ensemble with the same value of  $U$ . The description of each ensemble tells us what application it is intended for. However, we often cheat and use whatever ensemble is convenient in calculations, regardless of the actual system we are trying to study. Why can we do that? The reason can be glimpsed in example 3.1: Fluctuations from the ensemble average are typically very small. This applies to any quantity which is variable in an ensemble, so for example the fluctuations in temperature in a microcanonical ensemble, even though they are allowed since  $T$  is not held constant in this ensemble, are very small. It therefore makes very little difference in practice whether we hold  $U$  or  $T$  constant, or  $N$  or  $\mu$ , unless the system we are interested in is either very small or has some mechanism for amplifying fluctuations[4].

## ENSEMBLE CLASSIFIERS

Ensemble classifier generation methods using can be broadly classified into six groups that are based on (i) manipulation of the training parameters, (ii) manipulation of the error function, (iii) manipulation of the feature space, (iv) manipulation of the output labels, (v) clustering, and (vi) manipulation of the training patterns.

### Ensemble Classifier Generation by Manipulation of the Error Function

A group of ensemble classifier construction methods address this issue by augmenting the error function of the base classifiers. An error is imposed if base classifiers make identical errors on similar patterns. Negative correlation learning and is one such ensemble where all the individual networks in the ensemble are trained simultaneously and interactively through the correlation penalty terms in their error functions

### Ensemble Classifier Generation by Manipulation of The Feature Space

In another group of ensemble classifiers diversity among the base classifiers is achieved by manipulating the input feature space. Different feature subsets are used to train the base classifiers and. The random subspace ensemble classifiers perform relatively inferior to other ensemble classifiers.

### Ensemble Classifier Generation by Manipulation of the Output Labels

Ensemble classifiers can be constructed by manipulation of the output targets and. In class switching ensemble , each base classifier is generated by switching the class labels of a fraction of training patterns that are selected at random from the original training set[5].

## Properties of "good" ensembles

The following properties are considered desirable for a classical mechanical ensemble.

- Representativeness

The chosen probability measure on the phase space should be a Gibb's state of the ensemble, i.e., it should be invariant under time evolution. A standard example of this is the natural measure (locally, it is just the Lebesgue measure) on a constant energy surface for a classical mechanical system. Liouville's theorem states this measure is invariant under the Hamiltonian flow.

- Ergodicity

Once a probability measure  $\mu$  on the phase space  $\Lambda$  is specified, one can define the *ensemble average* of an observable, i.e., real-valued function  $f$  defined on  $\Lambda$  via this measure by

$$\langle f \rangle = \int_{\Lambda} f d\mu$$

where we have restricted to those observables which are  $\mu$ -integrable.

On the other hand, let  $x(0)$  denote a representative point in the phase space, and  $x(t)$  be its image under the flow, specified by the system in question, at time  $t$ . The *time average* of  $f$  is defined to be

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(x(t)) dt,$$

provided that this limit exists  $\mu$ -almost everywhere and is independent of  $x(0)$ .

The ergodicity requirement is that the ensemble average coincide with the time average. A sufficient condition for

ergodicity is that the time evolution of the system is a mixing. (See also ergodic hypothesis.) Not all systems are ergodic. For instance, it is unknown at this time whether classical mechanical flows on a constant energy surface are ergodic in general. Physically, when a system fails to be ergodic, we may infer that there is more macroscopically discoverable information available about the microscopic state of the system than what we first thought. In turn this may be used to create a better-conditioned ensemble[2].

## Ensembles in quantum statistical mechanics

Putting aside for the moment the question of how statistical ensembles are generated operationally, we should be able to perform the following two operations on ensembles  $A, B$  of the same system:

- Test whether  $A, B$  are statistically equivalent.
- If  $p$  is a real number such that  $0 < p < 1$ , then produce a new ensemble by probabilistic sampling from  $A$  with probability  $p$  and from  $B$  with probability  $1 - p$ .

Under certain conditions therefore, equivalence classes of statistical ensembles have the structure of a convex set. In quantum physics, a general model for this convex set is the set of density operators on a Hilbert space[2]. Accordingly, there are two types of ensembles:

- *Pure ensembles* cannot be decomposed as a convex combination of different ensembles. In quantum mechanics, a pure density matrix is one of the forms  $|\phi\rangle\langle\phi|$ . Accordingly, a ray in a Hilbert space can be used to represent such an ensemble in quantum mechanics. A pure ensemble corresponds to having many copies of the same (up to a global phase) quantum state.
- *Mixed ensembles* are decomposable into a convex combination of different ensembles. In general, an infinite number of distinct decompositions will be possible[5].

Thus, a quantum mechanical ensemble is specified by a mixed state in general. For example, one can specify the density operators describing microcanonical, canonical, and grand canonical ensembles of quantum mechanical systems, in a mathematically rigorous fashion.

The normalization factor required for the density operator to have trace 1 is the quantum mechanical version of the partition function.

We note here that ensembles of quantum mechanical system are sometimes treated by physicists in a *semi-classical* fashion. Namely, one considers the phase space of the corresponding classical system (e.g., for an ensemble of quantum harmonic oscillators, the phase space of a classical harmonic oscillator is considered). Then, using physical arguments, one derives a suitable "fundamental volume" for the particular system to reflect the fact that quantum mechanical microstates are discretely distributed on the phase space. From the uncertainty principle, it is expected this fundamental volume to be related to the Planck constant,  $\hbar$ , in some way[6].

## Operational interpretation

In the discussion given so far, while rigorous, we have taken for granted that the notion of an ensemble is valid a priori, as is commonly done in physical context. What has not been shown is that the ensemble *itself* (not the consequent results) is a precisely defined object mathematically[7]. For instance,

- It is not clear where this *very large set of systems* exists (for example, is it a gas of particles inside a container?)
- It is not clear how to physically generate an ensemble.

In this section we attempt to partially answer this question.

Suppose we have a *preparation procedure* for a system in a physics lab: For example, the procedure might involve a physical apparatus and some protocols for manipulating the apparatus. As a result of this preparation procedure some system is produced and maintained in isolation for some small period of time. By repeating this laboratory preparation procedure, we obtain a sequence of systems  $X_1, X_2, \dots, X_k$ , which in our mathematical idealization, we assume is an infinite sequence of systems. The systems are similar in that they were all produced in the same way. This infinite sequence is an ensemble.

In a laboratory setting, each one of these prepped systems might be used as input for *one* subsequent *testing procedure*. Again, the testing procedure involves a physical apparatus and some protocols; as a result of the testing procedure, we obtain a *yes* or *no* answer. Given a testing procedure  $E$  applied to each prepared system, we obtain a sequence of values  $\text{Meas}(E, X_1), \text{Meas}(E, X_2), \dots, \text{Meas}(E, X_k)$ . Each one of these values is a 0 (or no) or a 1 (yes).

Assume the following time average exists:

$$\sigma(E) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N \text{Meas}(E, X_k)$$

For quantum mechanical systems, an important assumption made in the quantum logic approach to quantum mechanics is the identification of *yes-no* questions to the lattice of closed subspaces of a Hilbert space. With some additional technical assumptions one can then infer those states are given by density operators  $S$  so that:

$$\sigma(E) = \text{Tr}(ES).$$

We see this reflects the definition of quantum states in general: A quantum state is a mapping from the observables to their expectation values[7].

## References

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